



Institut de Recherche
en Informatique de Toulouse
CNRS - Toulouse INP - UT - UT Capitole - UT2

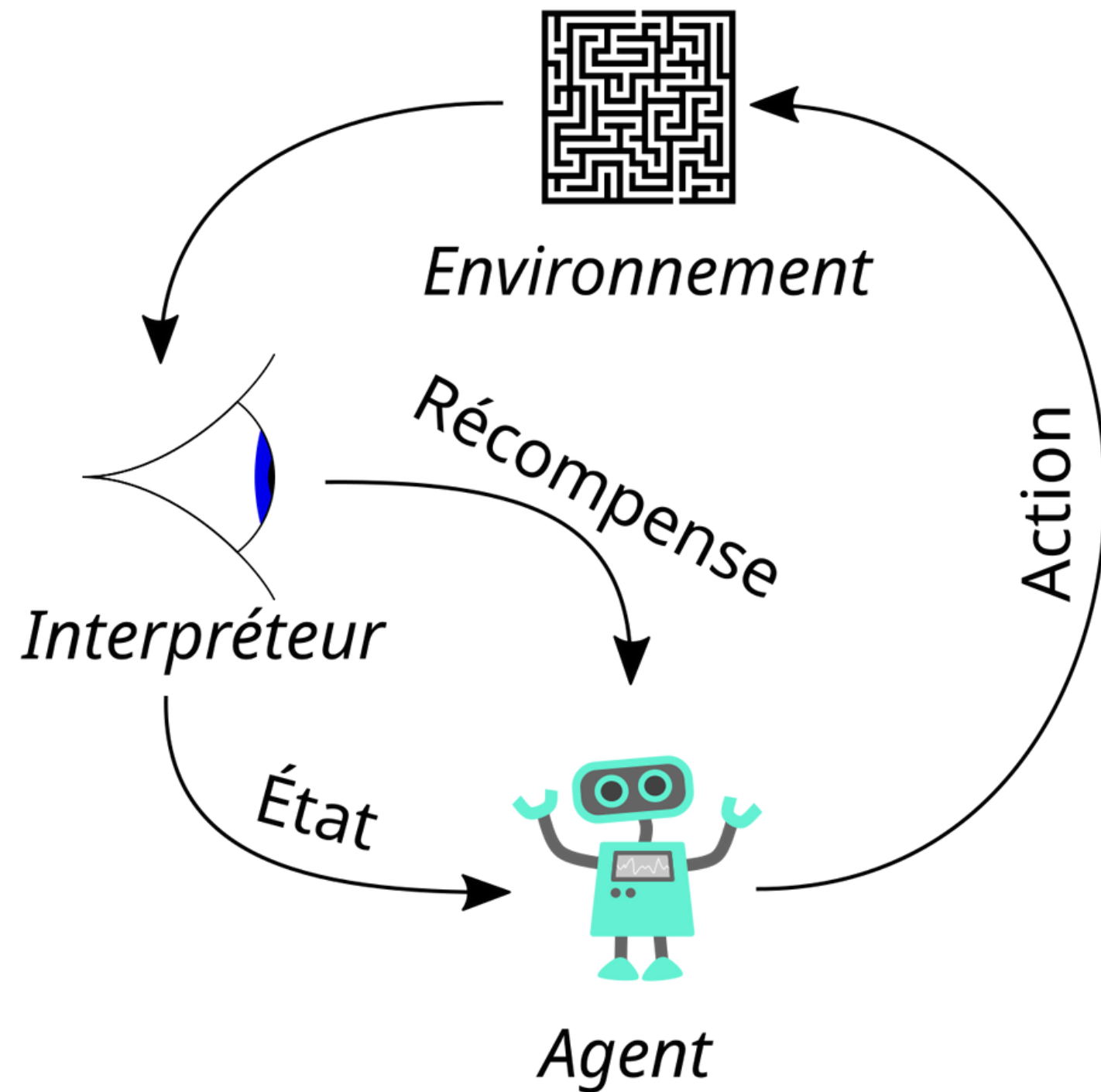
Sujet : étude de la capacité des abeilles à répondre à un problème d'apprentissage séquentiel.

Encadrante: Emmanuelle Claeys - Chercheuse IRIT

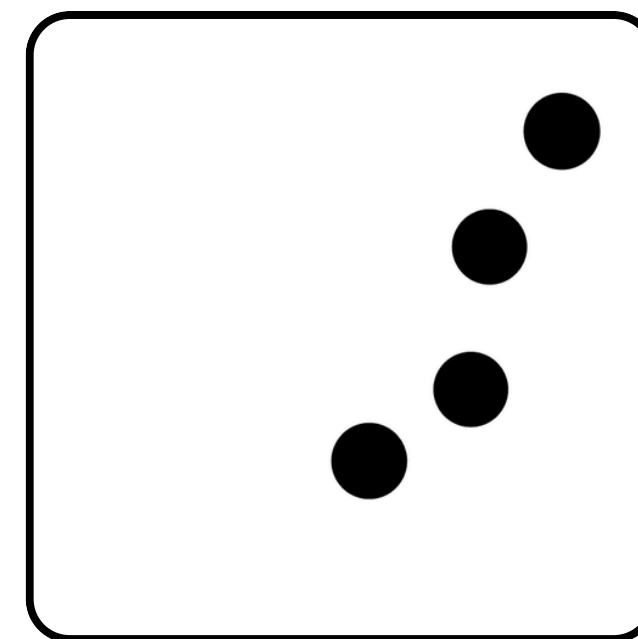
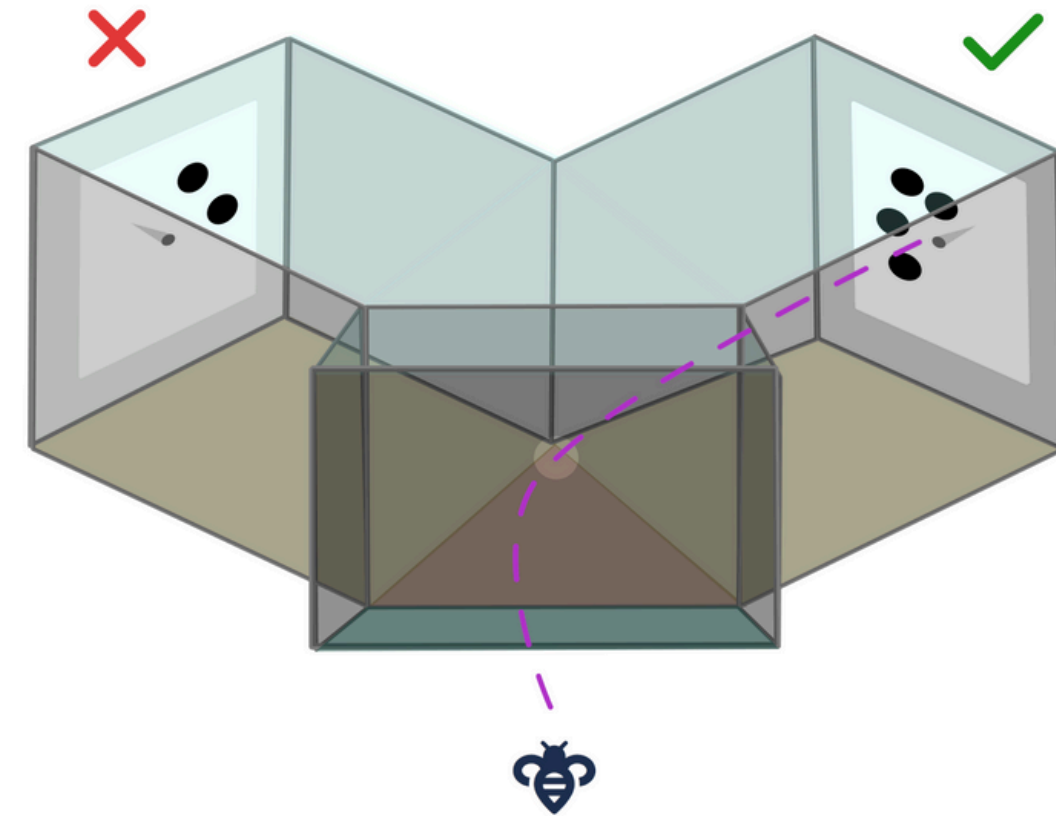
Aurore Avarguès-Weber - Chercheuse CBI

Elena Kerjean - Doctorante CRCA / CBI

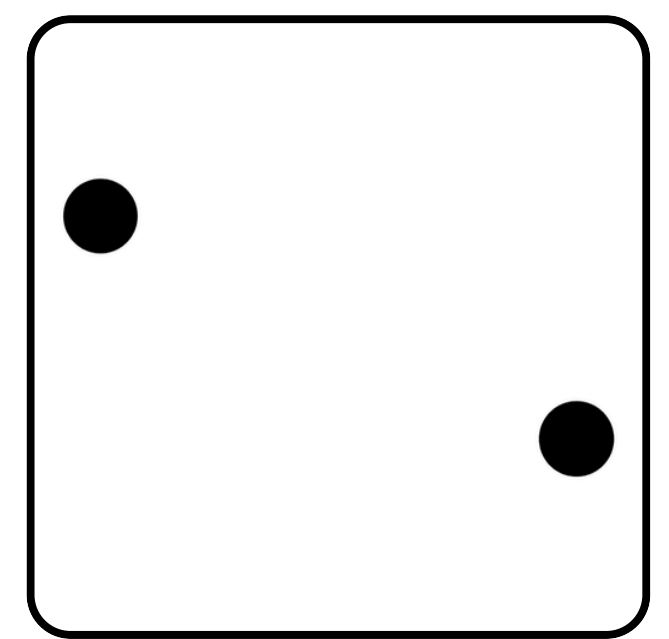
Multi-Armed Bandits



Apprentissage par renforcement



Récompense (4 points noirs)



Punition (2 points noirs)

Importation des données

BeesRLWithR

Dashboard

Upload Dataset(s)

Select Individual

Select Group

Select Dataset

Settings

Help

Information

Quit

Import a file

Browse...

No file selected

Imported Files

NB_Data_Summer_Australia_2024_06_03_2024...

Select

SM_Data_Summer_Australia_2024_14_02_2024...

Select

Number of individuals : 16

Number of choices : 640

Errors:

No errors found in the dataset

NB_Data_Summer_Australia_2024_06_03_2024_Number_bias.xlsx

Search:

id	trial	side	pair	correct
2	1	D	1	0
2	2	D	7	0
2	3	G	2	0
2	4	D	7	0
2	5	D	6	0
2	6	G	3	0
2	7	G	4	1
2	8	D	7	0
2	9	G	7	0
2	10	D	3	1
2	11	G	9	0
2	12	G	3	0
2	13	D	3	1
2	14	G	7	0
2	15	G	2	0

Showing 1 to 15 of 640 entries

Previous

1

2

3

4

5

...

43

Next

Dashboard

BeesRLWithR

Dashboard

Upload Dataset(s)

Select Individual

Select Group

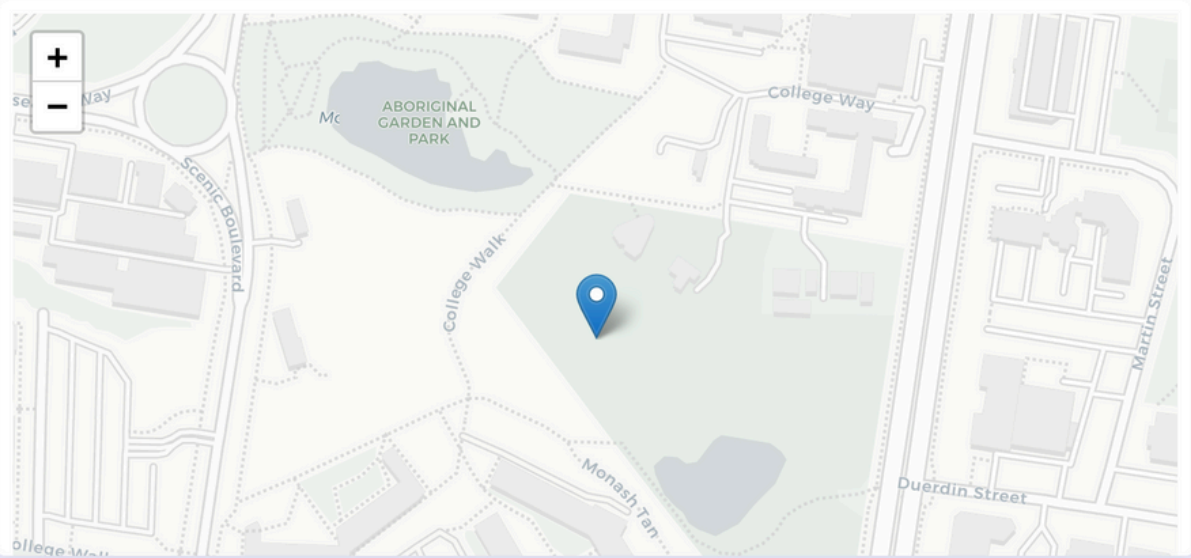
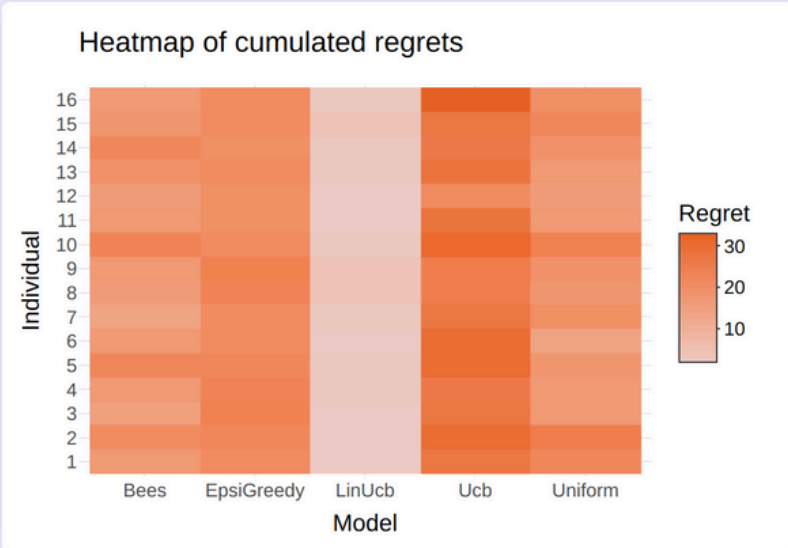
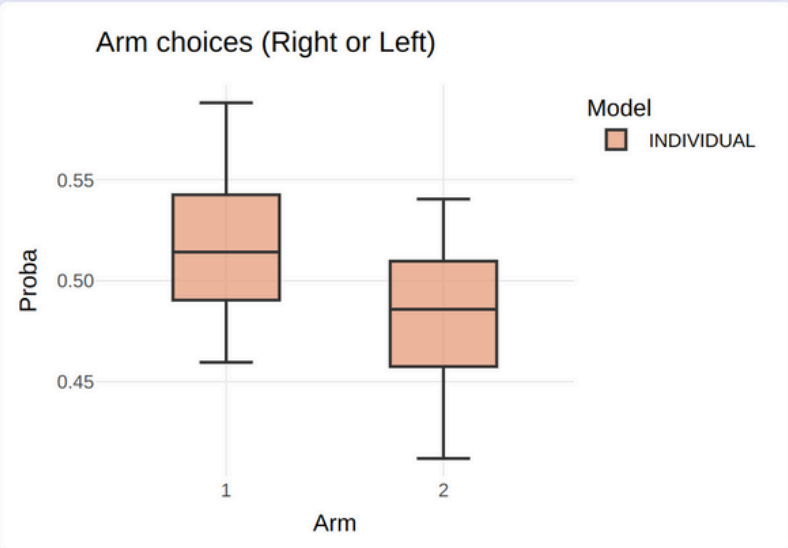
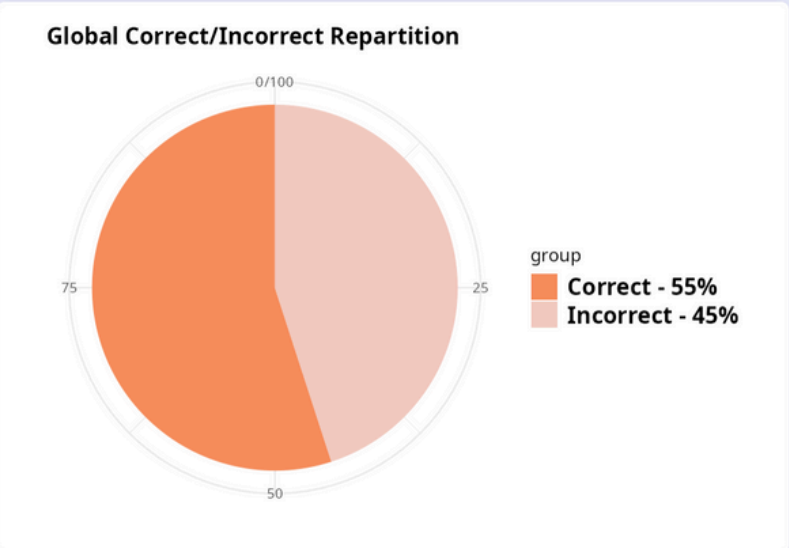
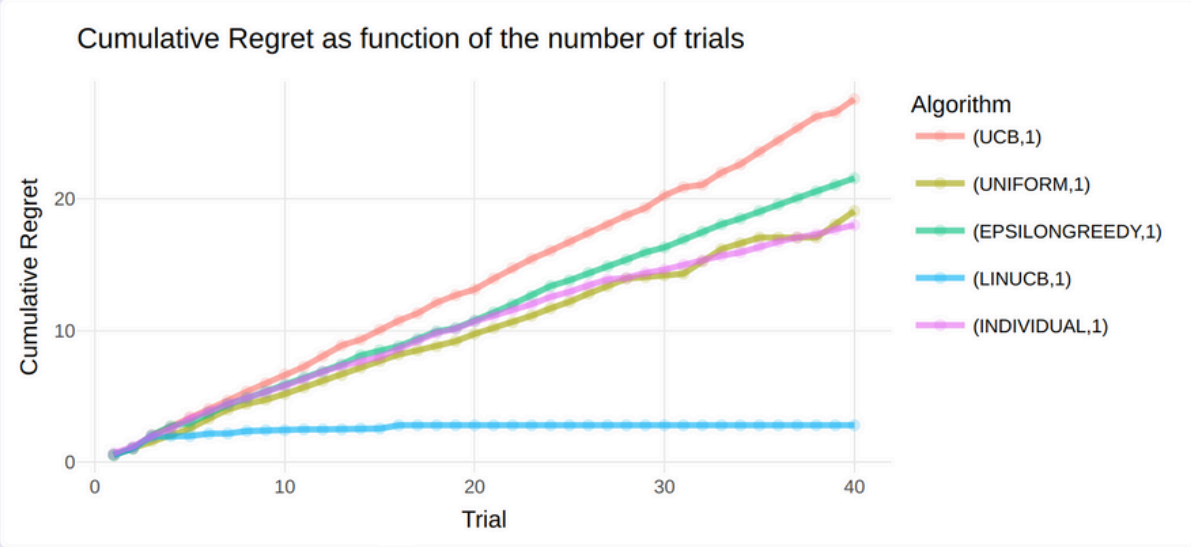
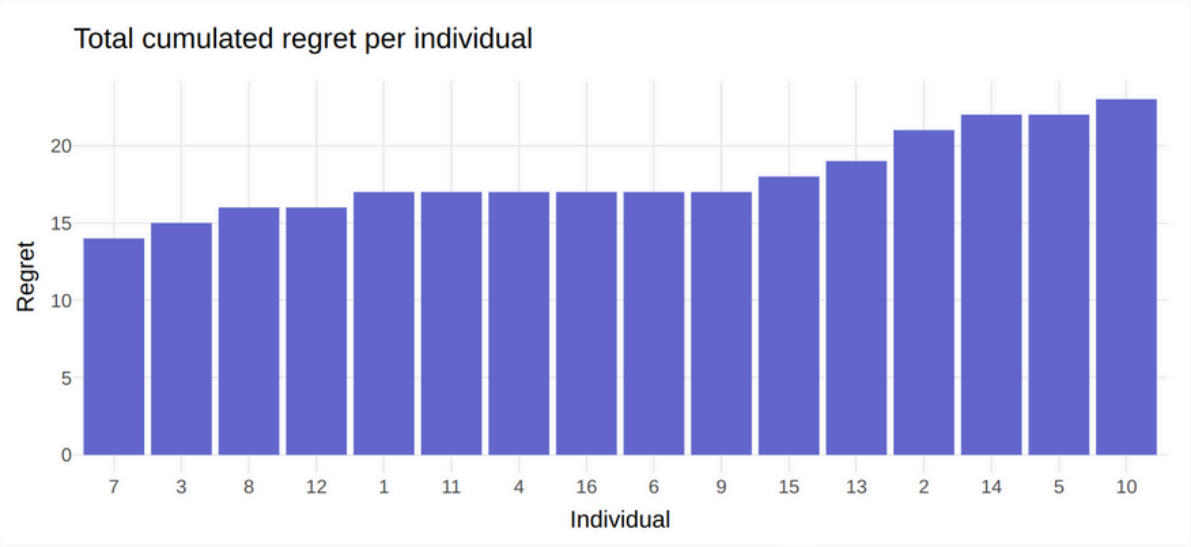
Select Dataset

Settings

Help

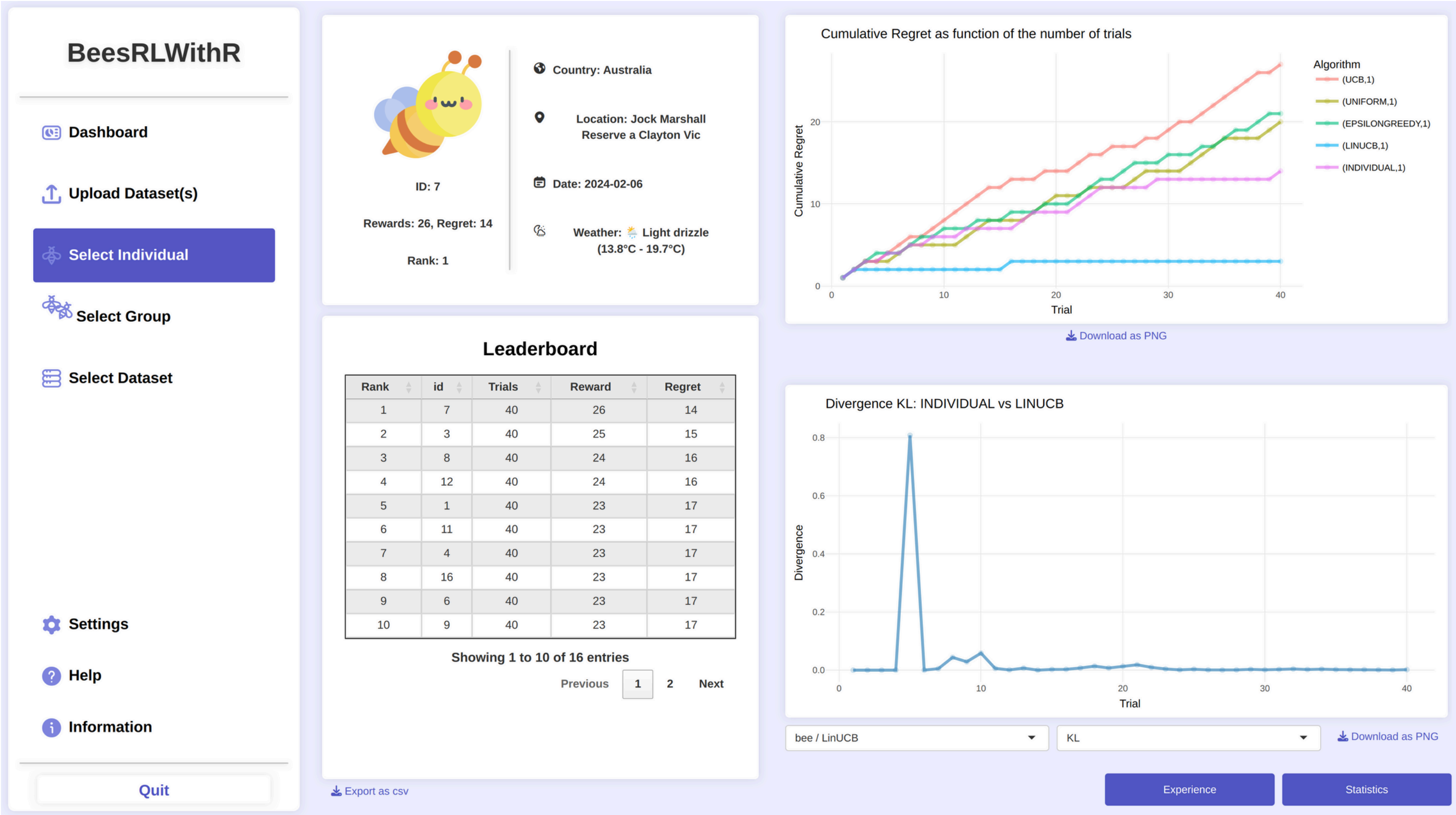
Information

Quit

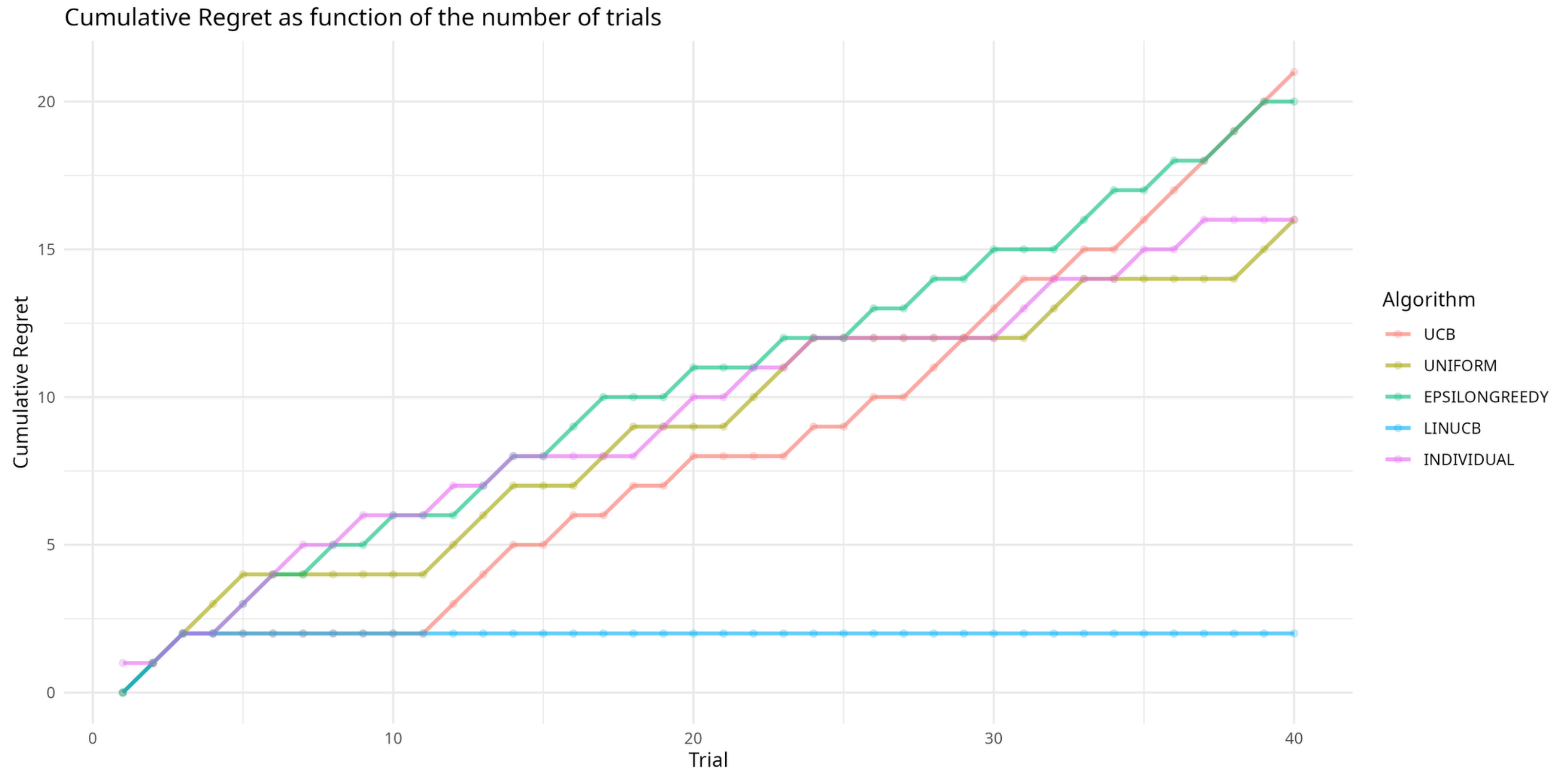


Rank	Model	Regret	Reward	LeftArm	RightArm
1	LinUcb	2.55	17.95	0.5	0.5
2	Uniform	9.98	10.52	0.49	0.51
3	Bees	10.36	10.14	0.48	0.52
4	EpsiGreedy	11.33	9.17	0.49	0.51
5	Ucb	13.78	6.72	0.46	0.54

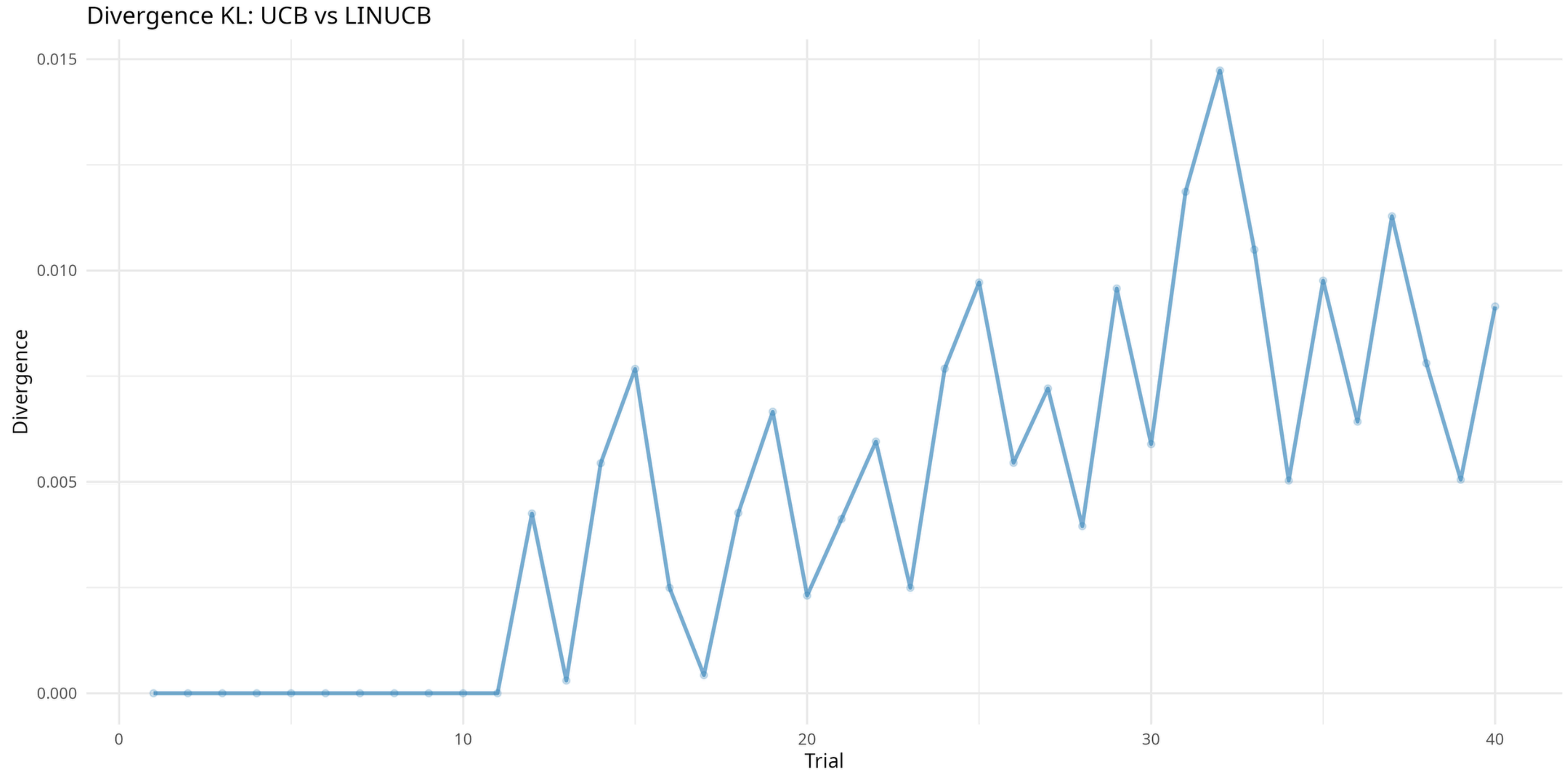
Analyse individuelle



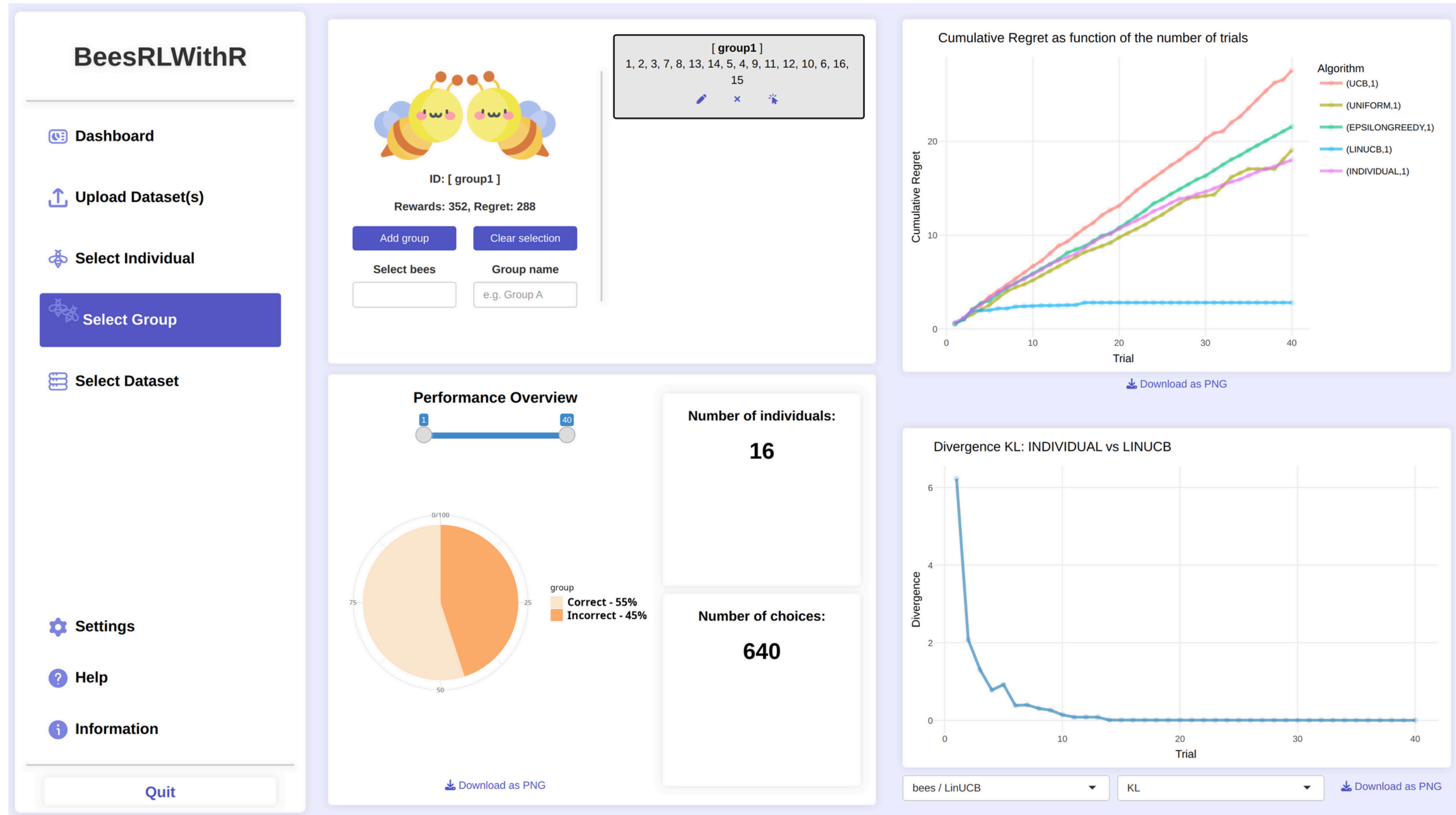
Tests statistiques (1)



Tests statistiques (2)



Analyse par groupe



Axes d'améliorations

- + Expériences scientifique
- + Modèles de RL pour la biologie
- + Performance
- + Correction de bugs
- + Documentation